

Progress Report: Sprint 10

CS.028 Optimizing Research Software Codes - 15 March 2026

Kathryn Butler, Michael McAllister, Joseph Schaab

Summary

- Major UI improvements including time logging and progress bars.
- Rerouted all warnings to warning_log.txt to address later.
- Completed Winter Term Retrospective, Video Demo, Peer Reviews.

Sprint Work Log

UI Improvements

Status: Ongoing (80% complete)

Author: Kathryn Butler

Reviewers: N/A

Source Link: [Commit 56](#)

Supporting Artifact:

```
t value for compat will change from compat='no_conflicts' to compat='override'. This is likely to lead to different results when combining overlapping variables with the same name. To opt in to new defaults and get rid of these warnings now use `set_options(use_new_combine_kwargs_defaults=True)` or set compat explicitly.
  obs = xarray.merge((obs, o))
C:\Projects\optimizing-research-software-codes\gnss_python-main\georinex\obs2.py:54: FutureWarning: In a future version of xarray the default value for compat will change from compat='no_conflicts' to compat='override'. This is likely to lead to different results when combining overlapping variables with the same name. To opt in to new defaults and get rid of these warnings now use `set_options(use_new_combine_kwargs_defaults=True)` or set compat explicitly.
  obs = xarray.merge((obs, o))
Processing started...
Using 8 cores to calculate positions.
If you encounter memory errors, reduce num_cores and try it again or disable parallel
C:\Users\Kated\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.12_qbz5n2kzffr8p0\LocalCache\local-packages\Python312\site-packages\umpy\_core\fromnumeric.py:57: FutureWarning: 'DataFrame.swapaxes' is deprecated and will be removed in a future version. Please use 'DataFrame.transpose' instead.
    return bound(*args, **kwargs)
```

Original Output

```
10:03:39 - =====  
10:03:39 - GNSS RINEX Processing Started  
10:03:39 - =====  
10:03:39 - Loading config...  
10:03:44 - Large obs version 3 file detected!  
10:03:44 - Converting obs version down to 2..  
  
10:05:31 - opening 178.12244 MByte SEPT2620.250  
10:05:42 - Processing RINEX files...  
10:05:50 - Calculating positions using 8 cores...  
Calculating satellite positions: 100%|██████████████████████████████████████| 32/32 [11:44<00:00, 22.02s/it]  
Building output rows: 100%|███████████████████████████████████████████████████████| 1749600/1749600 [04:37<00:00, 6298.83it/s]  
10:22:35 - Processing completed.  
10:22:35 - Total runtime: 1136.80 seconds (19 minutes).  
10:22:35 - =====
```

Improved Output

Notes: The original output contained several warnings that were sent directly to the console. It also did not let the user know that the program had begun running until about 30 seconds after it had begun. Regular users do not need to see these warnings and do not get feedback that the process was executed correctly.

I routed the warnings to a warning_log.txt file so they wouldn't clutter the console. I also created messages letting the user know that the process had begun and what step it was on. This allows the user to feel confident that the program is doing what it should be, as well as get a time estimate on how long it will take on their machine.

Rerouted Warnings to Warning_log.txt

Status: Done

Author: Kathryn Butler

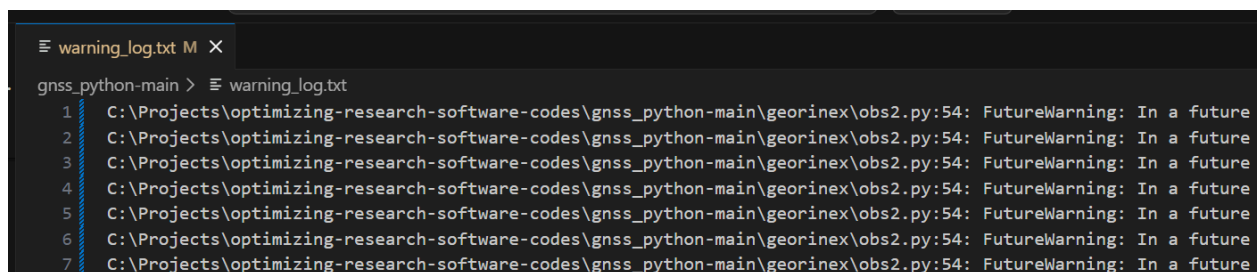
Reviewers: N/A

Source Link: [Issue 10](#)

Supporting Artifact:

```
t value for compat will change from compat='no_conflicts' to compat='override'. This is likely to lead to different results when combining
verlapping variables with the same name. To opt in to new defaults and get rid of these warnings now use `set_options(use_new_combine_kwa
defaults=True)` or set compat explicitly.
obs = xarray.merge((obs, o))
C:\Projects\optimizing-research-software-codes\gnss_python-main\georinex\obs2.py:54: FutureWarning: In a future version of xarray the def
t value for compat will change from compat='no_conflicts' to compat='override'. This is likely to lead to different results when combining
verlapping variables with the same name. To opt in to new defaults and get rid of these warnings now use `set_options(use_new_combine_kwa
defaults=True)` or set compat explicitly.
obs = xarray.merge((obs, o))
Processing started...
Using 8 cores to calculate positions.
If you encounter memory errors, reduce num_cores and try it again or disable parallel
C:\Users\kated\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.12_qbz5n2kfra8p0\LocalCache\local-packages\Python312\site-package
umpy\_core\fromnumeric.py:57: FutureWarning: 'DataFrame.swapaxes' is deprecated and will be removed in a future version. Please use 'Data
me.transpose' instead.
return bound(*args, **kwargs)
```

Original Console Output (Lots of Warnings!)



```
gnss_python-main > warning_log.txt
1 C:\Projects\optimizing-research-software-codes\gnss_python-main\georinex\obs2.py:54: FutureWarning: In a future
2 C:\Projects\optimizing-research-software-codes\gnss_python-main\georinex\obs2.py:54: FutureWarning: In a future
3 C:\Projects\optimizing-research-software-codes\gnss_python-main\georinex\obs2.py:54: FutureWarning: In a future
4 C:\Projects\optimizing-research-software-codes\gnss_python-main\georinex\obs2.py:54: FutureWarning: In a future
5 C:\Projects\optimizing-research-software-codes\gnss_python-main\georinex\obs2.py:54: FutureWarning: In a future
6 C:\Projects\optimizing-research-software-codes\gnss_python-main\georinex\obs2.py:54: FutureWarning: In a future
7 C:\Projects\optimizing-research-software-codes\gnss_python-main\georinex\obs2.py:54: FutureWarning: In a future
```

Warning_log.txt Warning Redirection

Notes: Rerouting things also may have made the program run a tiny bit faster. This could be within the margin of error, but stopping things from being sent to the console seems to have saved us about 40 seconds.

Initial run: calculate_positions = 513s, parse() = 741s

Optimized run: calculate_positions() = 480s, parse() = 704s

Developing Updated Math Logic for calculate_positions()

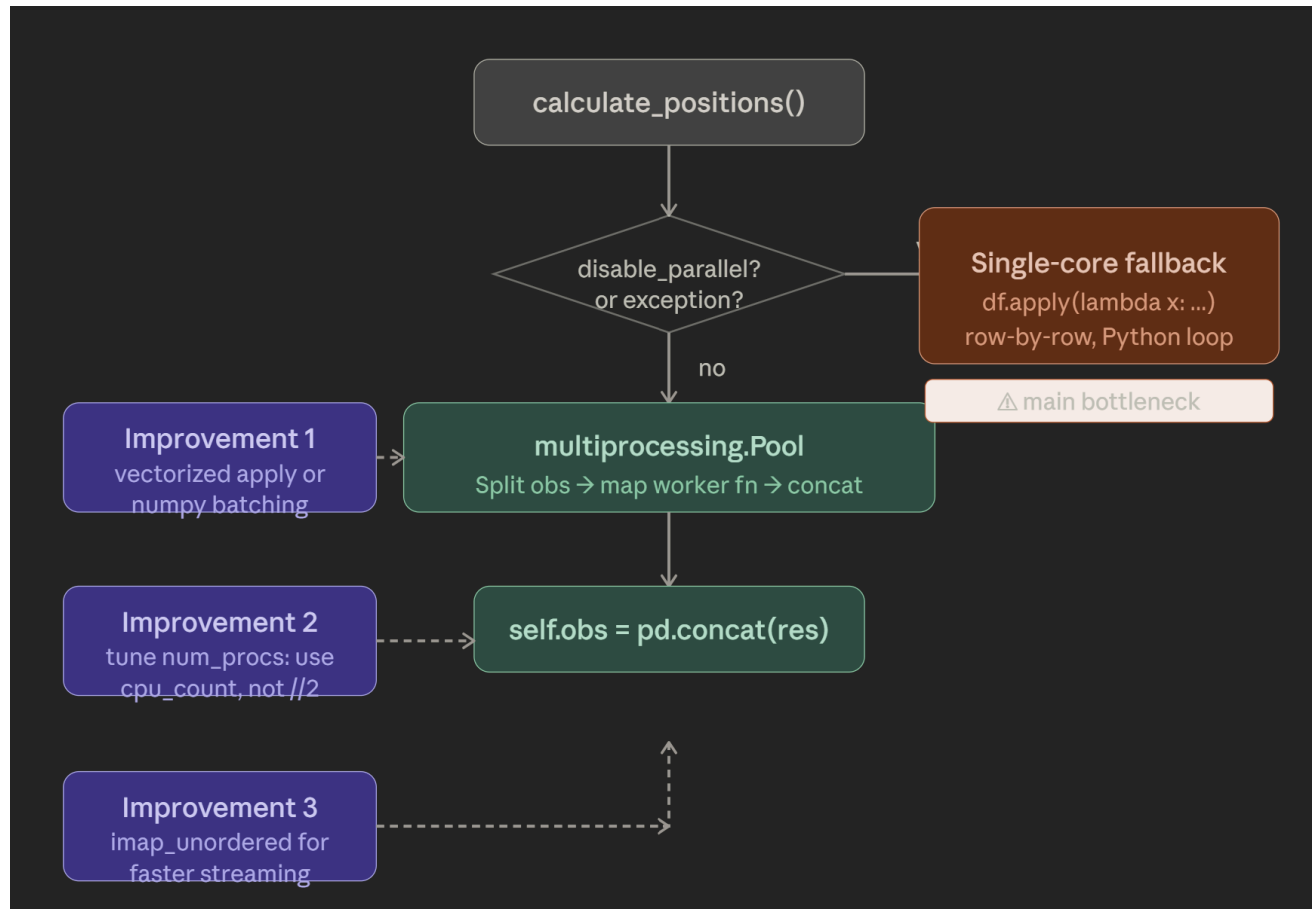
Status: Partial (40%)

Author: Joseph Schaab

Reviewers: N/A

Source Link: [OneNote - Solutions - calculate_positions\(\)](#)

Supporting Artifact:



Flowchart with potential improvements for calculate_positions()

Notes: Solution options have been drafted and commented on in the [OneNote](#). Still need to test solutions individually against each other to find the option that provides the most optimization. The solution that tests the best will be committed into the main repository.

Risk and Quality Tracking

Bug Count: 11 ([Link to issue query](#))

Top Bugs:

- [Treating Keys as Positions and Using Argsort](#)
 - **Severity:** High
 - **Owner:** Kathryn Butler
 - **Current Status:** Resolved
 - **Actions:** No further action needed. (Series.getitem treating keys as positions is deprecated. Look up how to solve it.)
- [Incorrect conversion from float to datetime](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Resolved
 - **Actions:** No further action needed. (Research into the line causing the problem. Determine the input causing the problem, and correct the datetime variable.)
- [Log.txt hardcoded](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Resolved
 - **Actions:** No further action needed. (Log.txt seems to be announcing a preset number of cores to calculate. Figure out why it is set to only be 8 cores. Find out if you can manually adjust the core count to be something other than just 8, then add the text to the function dynamically.)
- [Inconsistent timing data](#)
 - **Severity:** High
 - **Owner:** Kathryn Butler
 - **Current Status:** Ongoing
 - **Actions:** Create a smaller test data pool, run the tests multiple times to determine if the size of the file increases the likelihood of failed parsing attempts.
- [Xarray.merge conflict](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Ongoing
 - **Actions:** Set compat explicitly to something other than 'override', then test output for clarity. (Previous goal: Research what "compat" does, and either set it explicitly or use the new combine_kwargs_defaults.)

Next Sprint Plan

- Finish updating UI
 - Modify progress bar to have a more descriptive sidebar. Add additional “beginning processing” notifications that appear right as the program starts (or as close as we can get it). Get additional feedback from our project partner about the style and any changes that would be beneficial. Source of truth: most recent [Meeting Notes](#)
- Solve more bugs
 - Specifically the warning statements need to be resolved. Find updated libraries that contain the functionality the program needs. Also stop the converting to obs and nav version 2, we got feedback that the functions are deprecated. Source of truth: [Issue pool](#)
- Create reproducible testing data from existing dataset
 - Generate a smaller dataset to reproduce runtime inconsistencies. Establish baseline timing benchmarks and document results. Commit dataset to repository. Source of truth: [Issue 4](#)

Team Process Reflection

Now that we have assumed that Michael is not participating in the project anymore, task distribution is much simpler. We were able to divide work between `parse()` and `calculate_positions()`, and have made better progress than we have in previous sprints.

Some blockers we are experiencing are the fact that since many of the functions have been optimized, it is difficult to find further improvements. To mitigate this, we have shifted our focus slightly to wrapping up the UI of the program so users can find it intuitive to use, and future developers have an easier time understanding how it works.